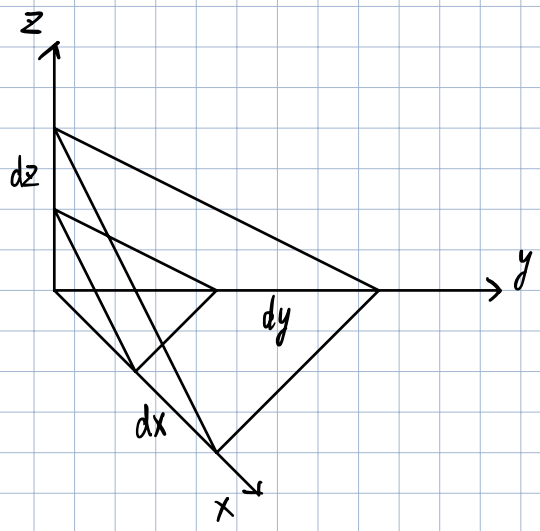


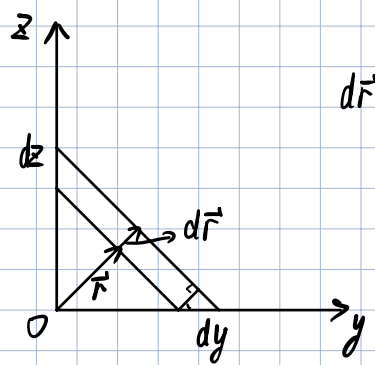
$$\phi_1 = (\vec{k}\cdot\vec{r}_1 - \omega t_1) = \phi_2 = (\vec{k}\cdot\vec{r}_2 - \omega t_2)$$

$$\vec{k}\cdot(\vec{r}_2 - \vec{r}_1) = \omega(t_2 - t_1).$$

$$\vec{k}\cdot\Delta\vec{r} = \omega\Delta t = k_x dx + k_y dy + k_z dz.$$

$$= \omega dt \qquad = k\,dy\cos\theta$$

$$v_{py} = \frac{dy}{dt} = \frac{\omega}{k_y} = v_p / \cos\theta.$$



$$d\vec{r} = dy\cos\theta.$$